# Sacred Texts

Electrodynamics as the derivative of the field equations, read through the card between

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The card the reviewers hold the game to, and the press that sets it.

"Should such monkey-patched code exist, we can use that as the Temple that stores our sacred texts."

## Reading note

Two episodes of the middle card are set here as mathematics. In the first, the accomplice, a reviewer holds the game to a canon of received results and the game answers each demand with a located reference and a calibration. In the second, the force, the answers become a publication: a register, a counter, a page trace, and a press verdict. This card performs both.

Traceless gravity yields its cosmological constant as an integration constant, and at a fixed value of that constant the affine term has an exact derivative, a constant multiplier. The scalar residual of a cubic action has an exact linearization with an exact remainder. Newton's update and bisection certify the same reporting cell to twenty-four places. Two readers that look at one record through a carrier agree, or change the calculation, according to how they account for the slot: the reader that deducts it reproduces the reading, and the reader that keeps it turns Newton's update into a map whose fixed point near the reading repels, so that no orbit of it from a rational seed converges.

Part I is the canon, kept as a register of dependencies, and the rule by which a received text is admitted. Part II brings the results above to one problem: the proposal, made on the numerical card, that the residual of electrodynamics is the derivative of the residual of the field equations, carried across a map between field spaces. That map and the operator identity are not established here (Section 10). Part II records what the scalar case and the carrier supply toward them, and what remains owed.

Each statement is a calibrated reading, and its label names the reference against which it is calibrated. *Proved here*: the proof is on this card. *Standard*: cited. *Measured*: a count, a definition, or a kernel-checked equality taken from the middle card, or the reported outcome of a run on it; the middle card is named only in Appendix A. *Recorded*: a chain of steps whose every premise is proved here, cited, or measured. *Proposed*: the continuation the knower carries, stated and not derived. A proposed statement asserts that an isomorphism exists, between the numbering of the construction's descriptions by the natural numbers and the description of what it is identified with, that the machine can exploit as a computation. Both sides are finite records: the reading is a computable number, and the reference is a value with its uncertainty or a stated structure. The statement is therefore one the machine can evaluate, a case that could be false, and the comparison that would revise it is that evaluation. What no evaluation supplies is that the reference's finite record stands for what it records, as a finite halt stands for its limit. *Predicted*: a number stated before the run that will decide it, with the run named and the statement sealed (Appendix B); a prediction is neither a step nor a proposal. The sealed predictions stand there unchanged; their outcomes are reported beside them and cited in the body wherever a prediction is used.

The companion cards are cited by section title, and single results by their printed number: the numerical card (*Four Models and a Common Fourth-Order Closure*, build of 29 October 2026), the geometric card, the Quarter, the bent card, the cheat sheet (*A Knower's Guide to Orbital Mechanics*) and the patter (*The Commutator of Representation*).

### In five lines

1. A received text counts through the fact it lets you compare. *Proved here* as a rule of admission (Section 1).

2. On the traceless sector the affine term of the field equations is an increment: what the integration adds to the traceless operator. At fixed integration constant its derivative is the

constant multiplier; for unconstrained variations a scalar-curvature term remains. *Proved here* (Section 6).

3. The first derivative of a cubic action is one object read two ways; in the scalar case its remainder is exact, and at the reading it squares to the discriminant. *Proved here* in the scalar case, cited in the field case (Section 7).

4. The search by the first variation and the search by its derivative share one jar and one point, to every printed place. *Proved here* and *measured* (Section 8).

5. Read through the card between, the reader that deducts the slot reproduces the reading. The reader that keeps it changes the inferred step from $(13, 18, 2)$ to $(14, 19, 3)$, whose fixed point above the pole repels, so that no orbit from a rational seed converges to it. *Proved here.* In the run of Appendix B that reader returned no reading before its guard stopped it. *Measured*; the sealed prediction of a small displacement is not borne out (Section 9).

# 1 The initiation

A science is taught from the front of a room: someone who solved a hard problem in the seat you are sitting in explains how, and either you see the trick or you do not. What is handed over in that room is a text. The reviewer of Part I holds the game to such texts, and the game needs a rule for admitting them that does not reduce to devotion.

**Definition 1.1** (Three standings of a text)**.** A received text stands in one of three ways.

(i) An *inside joke*: a statement true because it is checked by the apparatus that states it, with nothing compared. Reflexivity of equality is the example: $T = T$.

(ii) A *Pythagorean* text: a statement together with the belief that travels with it, cited and used, not shown on the card that uses it.

(iii) A *triangle*: a statement that hands back a fact which can be compared with another fact, and so can slip.

*Measured*: the three standings are the middle card's three ways of building the sacred texts of an initiation, and its reviewer's reply on whether devotion to a method has been mistaken for a method lowers the text to its triangle, the only standing that returns a fact. A standing records how a text is admitted; a calibration records how a statement that uses it is to be read.

**Proposition 1.2** (Rule of admission)**.** *A text admitted to this card is used only through what its standing returns: an inside joke through the check that states it, a Pythagorean text through the single place it is admitted, and a triangle through the fact it returns. What is derived at the point of admission is afterwards used through that derivation, not through the text again. No text is used through the belief that travels with it.*

*Proved here*, by inspection of every citation in this card against the table of Section 2; it is a rule the card keeps, and the table is its record.

# Part I. The accomplice

## 2 The canon

The table is a register of dependencies: every received text the card stands on, with its standing, the hypotheses under which it is applied and the object it is applied to, and the consequence it has here. A triangle names the fact it returns; a Pythagorean text names its point of admission.

| Text, *standing* | Hypotheses; object applied to | Consequence here |
|---|---|---|
| Reflexivity, $T = T$<br>*inside joke* | none;<br>any record | the coin of every record; checked, not compared |
| Symbol identity, value identity, choice (the axiom, as a check bills it)<br>*inside joke* | the checker's own rules;<br>every receipt | the three receipts a check can bill;<br>the printed reading bills the first only (Section 8) |
| Einstein's field equations [8]<br>*Pythagorean* | a Lorentzian metric with its Levi-Civita connection;<br>the region of Section 6 | the equations $G + \Lambda g = 0$ whose derivative is taken (Section 6) |
| Palatini's first-order variation [15]<br>*triangle* | densitized inverse metric and independent torsion-free connection;<br>the first-order action | the action is a cubic polynomial and its fourth variation vanishes: numerical card Prop. 2.1, Thm. 2.4 |
| Unimodular gravity [17, 11, 19]<br>*triangle* | fixed volume density;<br>the traceless equations | the cosmological constant as an integration constant: numerical card Prop. 2.2, restated as Prop. 6.1 |
| Contracted Bianchi identity, $\nabla^\mu G_{\mu\nu} = 0$<br>*Pythagorean* | Levi-Civita connection of $g$;<br>a connected solution of $E(g) = 0$ | $R$ is constant, $R = 4\Lambda$ (Prop. 6.1);<br>used afterwards in Prop. 6.3 |
| Dirac's equation and electrodynamics [7]<br>*Pythagorean* | the bare theory on a fixed background;<br>the model whose residual is proposed as the derivative | the target of the proposal (Section 10) |
| Gravity as the square of gauge theory [1, 2, 3]<br>*Pythagorean* | perturbative amplitudes;<br>the shape of the proposal | the shape of the proposal, and no more (Section 10) |
| Fréchet and Gâteaux derivatives coincide for bounded cubics<br>*triangle* | bounded multilinear forms on a Banach space;<br>the cubic actions | numerical card Prop. 2.5; Prop. 7.1 here, with its remainder |
| Density of smooth functions in Sobolev space [13]<br>*Pythagorean* | first-order Sobolev space on a domain;<br>the first variation | pointwise and Sobolev first variations are one functional (numerical card, *The derivative of the modified field equations*) |
| Polynomial approximation [18, 4]<br>*Pythagorean* | continuous functions on a closed interval, uniform norm;<br>the trapped solution of Section 10 | convergence of values, not of derivatives (Section 10) |

| Text, *standing* | Hypotheses; object applied to | Consequence here |
|---|---|---|
| Newton's method<br>*triangle* | $r' \neq 0$ on the jar;<br>the residual $r$ | the exact error law, Prop. 8.3 |
| Bisection<br>*triangle* | a continuous $r$ with a sign change;<br>the jar | the halving ledger and its certificate, Prop. 8.4 |
| Herbrand's theorem [12]<br>*Pythagorean* | a first-order language, after Skolemization;<br>its ground instances | the space a refutation search ranges over (Section 9) |
| The fifth busy beaver [16]<br>*Pythagorean* | five-state, two-symbol Turing machines;<br>the name of the meter's bound | the name of a bound, and a proposed correspondence (Section 9) |
| Choice implies excluded middle [6]<br>*Pythagorean* | intuitionistic set theory, choice for all families;<br>the family of admissible slots | the slot of a carrier read as a choice (Section 9; the Quarter, *The quarter is a choice*) |
| Renormalization [10]<br>*Pythagorean* | perturbative electrodynamics with a regulator;<br>its divergent loop integrals | finite predictions for specified quantities (Section 4) |
| The revised SI [5]<br>*Pythagorean* | the defining constants fixed exactly;<br>the units of the record | $c$, $h$, $e$ exact, and the caesium second (Section 11) |
| The recommended fine-structure constant [14]<br>*triangle* | the 2022 adjustment;<br>the reading | 137.035999177(21), compared with the reading in Section 11 |
| The electron's anomalous moment [9]<br>*triangle* | the measured $g-2$ and its theory;<br>the reading | excludes the reading as a general value of the constant (numerical card, *The comparison with the measured constant*) |

Of twenty texts, two are inside jokes, seven are triangles and eleven are Pythagorean. Each Pythagorean text is admitted at one named place. What is derived there is afterwards used through the derivation: the constancy of $R$, obtained from the Bianchi identity in Prop. 6.1, is used again in Prop. 6.3 without the identity being admitted a second time. A reader who refuses a Pythagorean text therefore knows which line of Part II falls, and which later lines fall with it. *Recorded.*

**The reading in its three standings.** The reading this card records stands in all three ways at once, one part each. That the construction is checked, and that the check halts and returns zero, is an inside joke: the apparatus that states it checks it, and nothing is compared. It is the whole of what a reader needs, and each completed check is a halt observed on a machine. *Measured.* The conventions the reading is made under are Pythagorean: which comparison is named TRUE, which way a crossing runs, and which root is reported. Each is fixed once against a standard internal to the construction (the click, the order of the record, and the larger growth rate) and used at that place only, as the SI fixes the speed of light. The comparison with the recommended fine-structure constant [14] is the triangle, the only standing that returns a fact, and it slips by $1.80 \times 10^{-4}$, which the table of Section 11 carries into the impedance of free space. *Proposed*, as a classification of the reading; the slip is *Standard* against the cited value.

## 3 The thirty-six demands

The reviewer's demands are not reprinted here: the geometric and numerical cards each carry them in their chapters on the record and the press check (*The record and the press check*; *The record, the thirty-six replies, and the press check*), thirty-six replies, one per review page, each with a located reference and a calibration. What this card adds is the rule of Proposition 1.2 applied to the replies themselves. A reply that answers a demand with a proposition proved on its card returns a triangle. A reply that answers with a citation returns a Pythagorean text and names its point of admission. A reply that answers with the reading handed over by name, where the derived reading says less, is a proposal, and the numerical card's record lists those rungs. *Recorded*, from the two cards.

*Remark.* The accomplice and the force are the reviewer and the press. Which of the companion cards plays which is reversible, and no card is an instance of either; this card plays both, Part I as the account and Part II as the publication, in that order.

## 4 The canon all the way down

The canon of Section 2 lists the texts the derivation cites. A proof also stands on texts it never cites: the models that must hold for the proof to exist at all. Read from the top, each layer stands on the one below.

| Layer | What is assumed | Standing |
|---|---|---|
| the proof | its steps check | triangle |
| the checker | its logic is consistent and it implements that logic | Pythagorean |
| the compiled run | compiled evaluation computes what the source says; the checker reads only the literals it writes | Pythagorean |
| the operating system | memory and the work counter behave as specified | Pythagorean |
| the instruction set | each instruction does what its manual says | triangle |
| digital logic | levels beyond thresholds are clean bits, by engineered margins | Pythagorean |
| semiconductor physics | band structure and the transistor | Pythagorean |
| the electron | it carries charge as electrodynamics says | Pythagorean |

The stack answers five questions, and evidence for one does not answer another:

(a) whether a formal term establishes its proposition (the checker's verdict on the proof);

(b) whether an implementation executes the operation its source names (the compiled run);

(c) whether a given machine completed it (an observed halt, and the work it counted);

(d) what physical model accounts for the machine (semiconductor physics and, beneath it, electrodynamics);

(e) what measurements support that model in the regime the machine operates in (device characterization, and the precision tests of Section 11).

*Measured*, for the two triangles, and as an answer to (c). Forcing one symbol, the truth $T = T$, is recorded at 4286.1 units of the checker's work counter per symbol, and about 1.222 million instructions per symbol, on two machines (an M2 and an M2 Ultra), under the checker at version 4.27.0-rc1, over eight paired runs. The figures are per-symbol values over that workload, not single counts; on the two machines they differ by 0.08 percent, and the elapsed times differ. The counter's noise floor on the same toolchain is 3016 units; the record does not state whether it was subtracted from the per-symbol figure. On the tested machines the count belongs to the computation and the clock to the hardware: the carrier brings the count and the universe brings the time.

**The loop at the bottom.** A proof that a description is the electron's is checked by a machine whose correctness assumes the electron behaves as described. That is the sacred text at the foot of the stack, a Pythagorean text admitted under Prop. 1.2 at the last row of the table. The circle is not vicious, and the reason is engineering. Digital logic is built with thresholds and noise margins: a level counts as a bit only when it clears its threshold by a margin, so that the logical outcome is the same across a wide range of microscopic conditions. That margin lets the carrier support computation without the computation resolving the carrier's theory: question (c) is answered while (d) and (e) stay open, and the run needs only coarse, well-tested features of electrodynamics, not its fine structure. The loop runs through measurement, question (e), not through the derivation, question (a). *Recorded.*

**Watch for division by zero.** The recurring question is what rule lets a calculation continue at a singular point, and what that rule preserves. It has three different answers on this card.

(i) *Totalized division.* The checker's language defines division by zero to be zero, so that division is a total function; identities such as $(x/y)\,y = x$ then carry the hypothesis $y \neq 0$. What is preserved is totality, not the identity. A reader that lands on the pair $(0, 0)$ reads the number zero, silently, and one of the middle card's readers through the card between does exactly that (Appendix B). *Measured.*

(ii) *Newton's vanishing denominator.* Newton's step divides by the derivative, which vanishes at a double root, and that reader's quadratic has one. There the update has no value: nothing continues the calculation, and a branch must be chosen. *Proved here*, by Prop. 9.3, whose denominator $B(Dz - P)$ vanishes at $z = P/D$.

(iii) *Renormalization.* Electrodynamics computed beyond its first order meets divergent loop integrals. Renormalization introduces a regulator, redefines the parameters of the theory (mass, charge and field normalization) and adds counterterms, so that specified physical predictions are finite and independent of the regulator as it is removed; those predictions agree with measurement to high precision [10]. *Standard.* What is preserved is the chosen set of predictions; nothing is divided by zero.

That these are one move, a representation declaring what happens at a singular point so that the computation can go on, is *proposed*; the three rules preserve different things. A proof that looks like electrodynamics should be read for where its representation makes such a declaration: there the derivation has a sacred text.

This unbends the card.

# Part II. The force

## 5 The operator

The numerical card proves that four actions (first-order vacuum gravity, fixed-Hamiltonian Schrödinger evolution, incompressible flow, and bare electrodynamics with spinors on a fixed background) share one derivative hierarchy, numerical card Thm. 2.4: an action $S$ of degree at most three, a residual $R = DS$ of degree at most two, a derivative $DR = D^2S$ linear in the perturbation, a constant third derivative, and a vanishing fourth variation. For gravity the variables are the densitized inverse metric $h^{\mu\nu} = \sqrt{-g}\, g^{\mu\nu}$ and a torsion-free connection $\Gamma$, and the action is a cubic polynomial in $(h, \Gamma)$, numerical card Prop. 2.1. *Standard*, cited; the proofs are there. The next section passes from these variables to the metric operator whose derivative is taken, and says what the passage keeps.

## 6 The affine term as an increment

Work on a connected four-dimensional region with a Lorentzian metric $g$ and its Levi-Civita connection. Write $\mathrm{Ric}(g)$ for the Ricci tensor and $R(g)$ for the scalar curvature, and define the *traceless operator*

$$E(g) = \mathrm{Ric}(g) - \tfrac{1}{4}\, R(g)\, g.$$

*Remark* (From the cubic variables to the metric operator). The passage is

$$(h, \Gamma) \xrightarrow{\text{(i)}} (g, \Gamma) \xrightarrow{\text{(ii)}} g \xrightarrow{\text{(iii)}} E(g).$$

(i) is a change of variables: in four dimensions $\det(h^{\mu\nu}) = \det(g_{\mu\nu})$, so $g^{\mu\nu} = h^{\mu\nu}/\sqrt{-\det h}$, and the map is invertible on Lorentzian $h$. (ii) is elimination by a stationary equation: the variation with respect to $\Gamma$ gives, in vacuum and without torsion, the Levi-Civita connection of $g$, which is substituted back. (iii) is restriction to a sector: on the unimodular sector the stationary equations are $E(g) = 0$, numerical card Prop. 2.2. After (ii) the dependence on $g$ is no longer polynomial: the Levi-Civita connection involves $g^{-1}\partial g$, and the Ricci tensor is rational in $g$ and its first two derivatives. The degree bound of numerical card Thm. 2.4 therefore holds in $(h, \Gamma)$ and does not transfer to $E$ as an operator on metrics. *Proved here* for (i), by $\det(\sqrt{-g}\, g^{\mu\nu}) = (\sqrt{-g})^4 \det(g)^{-1} = \det g$; *standard*, cited, for (ii) and (iii).

**Proposition 6.1** (The affine term is the increment)**.** *For every $g$,* $\mathrm{Ric}(g) = E(g) + \frac{1}{4}R(g)\, g$. *If $E(g) = 0$, then $R(g)$ is a constant $4\Lambda$ and*

$$\mathrm{Ric}(g) - E(g) = \Lambda\, g, \qquad G(g) + \Lambda g = 0,$$

*where $G = \mathrm{Ric} - \frac{1}{2}Rg$ is the Einstein tensor [8]. The affine term $\Lambda g$ is what the integration adds to the traceless operator, and $\Lambda$ is not a coefficient of the action used here.*

*Proof.* The first identity is the definition of $E$. The contracted Bianchi identity gives $\nabla^\mu G_{\mu\nu} = 0$. If $E = 0$ then $G = \frac{1}{4}Rg - \frac{1}{2}Rg = -\frac{1}{4}Rg$, so $0 = \nabla^\mu G_{\mu\nu} = -\frac{1}{4}\partial_\nu R$, and on a connected region $R$ is a constant, written $4\Lambda$. Then $\mathrm{Ric} = \Lambda g$ and $G = \Lambda g - 2\Lambda g = -\Lambda g$. □

*Proved here*, from the Bianchi identity (*standard*). On the unimodular sector the stationary equations of the cubic action are exactly $E(g) = 0$ with $\Gamma$ the Levi-Civita connection, numerical card Prop. 2.2, so the proposition is the field equations of this card's gravity with the constant supplied as a residue.

**Proposition 6.2** (The derivative of the increment is the multiplier)**.** *At fixed $\Lambda$ the map $g \mapsto \Lambda g$ is affine, and its Fréchet derivative in any normed space of symmetric tensors is $h \mapsto \Lambda h$, with zero remainder.*

*Proof.* $\Lambda(g + h) - \Lambda g - \Lambda h = 0$. □

*Proved here.* The derivative of the full field equations is a different object, because $\Lambda$ is fixed by the solution and not by the action. The next proposition says what the traceless operator and the scalar curvature contribute beside the multiplier, and on which variations the multiplier is all the affine term contributes.

**Proposition 6.3** (The linearized increment)**.** *Let $E(g) = 0$ and $R(g) = 4\Lambda$, and let $k$ be a symmetric tensor. Then*

$$D\mathrm{Ric}_g[k] = DE_g[k] + \tfrac{1}{4}\, DR_g[k]\, g + \Lambda k.$$

*(a) For arbitrary $k$ the middle term is present.*

*(b) On variations with $DR_g[k] = 0$, the constrained sector, $D\mathrm{Ric}_g[k] = DE_g[k] + \Lambda k$, and the affine term contributes exactly the multiplier of Prop. 6.2.*

*(c) Along a differentiable family $g_s$ of solutions of $E(g_s) = 0$ with $g_0 = g$, $\dot g_0 = k$ and $R(g_s) = 4\Lambda(s)$, the affine term differentiates to $\dot\Lambda\, g + \Lambda k$ with $4\dot\Lambda = DR_g[k]$. The multiplier alone is the case of a fixed integration constant, $\dot\Lambda = 0$, which is case (b) on the family.*

*Proof.* Differentiate $\mathrm{Ric} = E + \frac{1}{4}Rg$ by the product rule: $D(Rg)[k] = DR_g[k]\, g + R(g)\, k$, and $\frac{1}{4}R(g) = \Lambda$. Case (b) is the substitution $DR_g[k] = 0$. For (c), differentiate $R(g_s) = 4\Lambda(s)$ and $\Lambda(s)g_s$ at $s = 0$. □

*Proved here.* An identification that proposes to match the derivative of the field equations must say which operator it matches: $DE_g + \Lambda$ on the constrained sector of (b), or the full $D\mathrm{Ric}_g$ of (a). Section 10 uses (b).

## 7 The first derivative, read two ways

The numerical card proves that for $S(Z) = a + b(Z) + c(Z, Z) + d(Z, Z, Z)$ with bounded multilinear forms on a Banach space, the Fréchet and Gâteaux derivatives coincide, $DS(Z)H = b(H) + 2c(Z, H) + 3d(Z, Z, H)$, and likewise for $R = DS$ and $DR$, numerical card Prop. 2.5. By density of smooth functions [13], the pointwise first variation and the Sobolev derivative are one bounded functional. *Standard*, cited. The scalar case is shown in full.

Let $S(y) = \frac{13}{3}y^3 - 18y^2 + 18y$. Its residual is

$$r(y) = S'(y) = 13y^2 - 36y + 18,$$

with roots $\alpha = (18 + 3\sqrt{10})/13$ and $\bar{\alpha} = (18 - 3\sqrt{10})/13$, and $d = \alpha - \bar{\alpha} = 6\sqrt{10}/13$.

**Proposition 7.1** (The scalar derivative, with its remainder)**.** *For all real $y, h$: $r(y + h) = r(y) + (26y - 36)h + 13h^2$. Hence $Dr(y)h = (26y - 36)h$ is both the Fréchet and the Gâteaux derivative, with remainder exactly $13h^2$. Moreover $(Dr(y))^2 = 360 + 52\, r(y)$, so $Dr(\alpha) = 6\sqrt{10} = 13d$.*

*Proof.* Expand. For the second identity, $(26y-36)^2 = 676y^2-1872y+1296 = 52(13y^2-36y+18)+360$. At $\alpha$ the residual vanishes and $26\alpha - 36 = 6\sqrt{10} > 0$. □

*Proved here*; the cheat sheet, Props. 2.1 and 2.2, proves the same. The identity $(Dr)^2 = 360 + 52r$ is a statement about one scalar quadratic; a field analogue would need its own object, a discriminant for an operator family, which this card does not construct. In the reported coordinate $x = \frac{324}{5}y$ the residual becomes, up to a positive factor, $325x^2 - 58320x + 1889568$, whose discriminant $944784000 = (9720\sqrt{10})^2$ is the square of its derivative at the upper root. The reported reading is

$$x_\alpha = \tfrac{324}{5}\,\alpha = \frac{972\,(6 + \sqrt{10})}{65}.$$

*Proved here*, since

$$\frac{324}{5} \cdot \frac{18}{13} = \frac{5832}{65} = \frac{972 \cdot 6}{65}, \qquad \frac{324}{5} \cdot \frac{3}{13} = \frac{972}{65}.$$

The coefficients are the numerical card's four-model quartic with $t = 5$, $c = 18$ and $s = 18$ (the numerical card writes $r$ for $s$), numerical card, *Neighbouring constructions*, where a scan of 412 neighbouring constructions finds no other within $10^{-3}$ of the reading.

**Proposition 7.2** (The quadratic in the Lorentz-factor form)**.** *Write the four-model quartic as $c(tx - s^2)^2 = t^3x^2$ and put $y = tx/s^2$ and $\beta = \sqrt{t/c}$. Then*

$$y_\pm = \frac{1}{1 \mp \beta}, \qquad y_+ y_- = \frac{1}{1 - \beta^2}, \qquad 1 - \beta^2 = \frac{c - t}{c}.$$

*With $t = 5$, $c = 18$: $\beta^2 = 5/18$, $1 - \beta^2 = 13/18$, so the leading coefficient 13 of $r$ is $c - t$, the product of its roots $18/13$ is $1/(1 - \beta^2)$, and $y_+ = \alpha$ gives the reading. As $\beta \to 1$, that is $t \to c$, the leading coefficient $t^2(c - t)$ vanishes, $y_+$ runs to infinity, and $y_-$ tends to $\frac{1}{2}$.*

*Proof.* $c(y - 1)^2 = ty^2$ gives $(y - 1)/y = \pm\beta$. The rest is substitution. □

*Proved here.* The exact content is the parametrization: under $y_\pm = 1/(1 \mp \beta)$ the product of the roots is $1/(1 - \beta^2)$, the form of the square of a Lorentz factor. Each factor $1/(1 \mp \beta)$ alone is the classical Doppler factor; what has the relativistic form is the product $y_+y_- = \gamma^2$. That $\beta$ is a velocity and $\beta = 1$ the speed of light is *proposed*. A map from $\beta$ to physical velocities that preserves the composition of boosts is a further task, and until it is given, quantities computed in the parametrization are properties of the parametrization.

## 8 Two searches, one jar, one point

Two searches find $\alpha$. They are built from different parts of the hierarchy, and that is their point.

**Definition 8.1** (The searches)**.** The *search by the first variation* is bisection: from walls $L < U$ with $r(L) < 0 < r(U)$, replace the wall whose residual has the sign of the midpoint's. It evaluates the sign of $r = S'$ and nothing else. The *search by the derivative* is Newton's update

$$N(y) = y - \frac{r(y)}{Dr(y)} = \frac{13y^2 - 18}{26y - 36},$$

which divides the first variation by its derivative. Where $Dr(y) \neq 0$, $N(y) = y$ exactly when $r(y) = 0$: the fixed points of the update are the zeros of the residual.

**Proposition 8.2** (The jar)**.** *$N(2) = 17/8$, $r(2) = -2 < 0 < 13/64 = r(17/8)$, and $2 > 18/13$, the vertex of $r$. The interval $[2, 17/8]$, in reported coordinates $[129.6, 137.7]$, contains $\alpha$ and no other root. It is the lower wall together with its first Newton image: the search by the derivative builds the jar the search by the first variation starts in.*

*Proof.* $N(2) = (52-18)/(52-36) = 34/16$. $r(2) = 52 - 72 + 18$ and $r(17/8) = \frac{3757}{64} - \frac{612}{8} + 18 = \frac{13}{64}$. Both walls lie above the vertex, where $r$ is increasing, so the sign change is a single root, the upper one. $\frac{324}{5} \cdot 2 = 129.6$ and $\frac{324}{5} \cdot \frac{17}{8} = 137.7$. □

*Proved here. Measured*: the middle card builds its jar walls from the lengths of its own tapes, not from the reader handed to its search, so every search the middle card runs, by whatever reader, starts in this jar. Whether it stays there depends on the reader (Prop. 9.6).

In the parametrization of Prop. 7.2 the walls sit at $\beta = 1/2$ and $\beta = 9/17$, and relativistic composition carries one to the other by

$$\frac{9/17 - 1/2}{1 - \frac{9}{17} \cdot \frac{1}{2}} = \frac{1}{25},$$

with rapidity $\frac{1}{2}\ln(13/12)$. *Proved here* as arithmetic in the parametrization. The jar is a finite boost below the asymptote $\beta = 1$, which every composition of boosts below one stays below: the penalty side of the constraint the asymptote meets exactly. *Proposed.*

**Proposition 8.3** (Exact error of the search by the derivative)**.** *For $e = y - \alpha$, $N(y) - \alpha = e^2/(2e+d)$. From any $y > \alpha$ the iterates decrease to $\alpha$ and the error is squared, up to the factor $1/(2e + d)$, at every step.*

*Proof.* $r(y) = 13e(e + d)$ and $Dr(y) = 13(2e + d)$, so $N(y) - \alpha = e - \frac{e(e+d)}{2e+d} = \frac{e^2}{2e+d}$. For $e > 0$ this lies in $(0, e/2)$. □

*Proved here*; numerical card Prop. 6.1. Since $2e + d = 2(y - 18/13)$, the same formula gives $N(y) \geq \alpha$ for every $y$ above the vertex.

**Proposition 8.4** (Certificate of the search by the first variation)**.** *After $n$ halvings of the jar the bracket has width $2^{-n} \cdot \frac{1}{8}$ and contains $\alpha$. With $S = 10^{24}$, at $n = 108$ both walls satisfy*

$$\left\lfloor S\,\tfrac{324}{5} L_{108} \right\rfloor = \left\lfloor S\,\tfrac{324}{5} U_{108} \right\rfloor = 137011290548979457087737300.$$

*Proved here* for the width and containment, by induction on the sign straddle; *standard*, cited, for the two floors, which the numerical card checks as integer inequalities $5KD \le 324SN < 5(K+1)D$ on the rational walls (numerical card, *A certificate obtained without either update*). Monotonicity of the floor then certifies the reading to twenty-four places:

$$x_\alpha = 137.011290548979457087737300\ldots$$

**Proposition 8.5** (One point)**.** *The search by the derivative, in the middle card's integer arithmetic at scale* $10^{24}$*, reaches the certified integer of Prop. 8.4 at its fifth turn. The search by the first variation reaches it within one hundred and eight halvings on this card's budget, and within the middle card's own budget of one halving per cell of its description, currently one hundred and six. Both walls and the derivative's point truncate to one integer.*

*Measured*: each line of both ledgers is checked by the kernel against the step before it, a checked ledger is proved equal to the fold of its step, the derivative's point is proved to lie in the final bracket above the vertex, and both walls are proved to print the same integer. The printed reading bills symbol identity only; the receipt that identifies the inferred integer with the box's ratio is a separate value identity, and it is not in the number. The numerical card reports the fifth turn (*The integer updates actually compared*).

*Remark* (What the two searches are)*.* The search by the first variation uses the sign of $S'$ alone: it locates a point where the first variation vanishes, a stationary point of the given action. It does not construct the action; the action is given, and the search finds where it is stationary. The search by the derivative uses $S'$ and $S''$: it is Newton's update, whose fixed points are the zeros of $r$, and it walks along the derivative of the residual, the object numerical card proposes as electrodynamics (*The derivative of the modified field equations*). That the first is the principle of stationary (least) action at work and the second the construction of electrodynamics is *proposed*. That they agree on the jar and on every printed place is *proved*. On that reading, the stationary point of the action and the fixed point of Newton's update coincide at the reading, as they must, both being the zero of $r$ in the jar, to twenty-four places; and the agreement is the scalar shadow of Section 10.

# 9 The operator inferred, and the card between

**Definition 9.1** (The inferred operator)**.** For positive integers $D, P, B$, on pairs of integers let

$$\Phi(p,q) = (Dp^2 - Pq^2,\ BDpq - BPq^2),$$

held at $(p,q)$ when its correction falls below a tolerance: writing $(p',q') = \Phi(p,q)$, when $|pq' - p'q|\ \sigma n < qq'm$ for a scale $\sigma$ and two further lengths $n, m$; on ratios, when $|p/q - p'/q'| < \tau = m/(\sigma n)$.

*Measured*: this is the middle card's inferred step. Its $D$, $P$ and $B$ are lengths of three of its tapes, $n$ and $m$ are two more, and the derivative the step divides by is never written down, only inferred from the shape of the map. The middle card computes on natural numbers, where subtraction truncates: $a \dot{-} b = 0$ when $b \ge a$. Its halting test writes the absolute value as $(a \dot{-} b) + (b \dot{-} a)$, which equals $|a - b|$ on natural numbers and is identically zero when read over the integers; the test relies on truncation.

**Definition 9.2** (Declared zero)**.** A correction is *declared zero* at resolution $\tau$ when it satisfies the stated stopping predicate, here $|p/q - p'/q'| < \tau$. The declaration does not assert that the exact correction vanishes.

**Proposition 9.3** (What the inferred operator computes)**.** *On ratios $z = p/q$, where $q \neq 0$ and $Dz \neq P$, $\Phi$ acts as*

$$G(z) = \frac{Dz^2 - P}{B(Dz - P)},$$

*whose fixed points satisfy $(B-1)Dz^2 - BPz + P = 0$. When $B = 2$ the map is Newton's update for $f(z) = Dz^2 - 2Pz + P$, and when moreover $D = 13$ and $P = 18$, $f = r$ and the map is $N$.*

*Proof.* Divide both components by $q^2$. A fixed point satisfies $Dz^2 - P = BDz^2 - BPz$. For $B = 2$, $f'(z) = 2Dz - 2P$ and $z - f(z)/f'(z) = (Dz^2 - P)/(2Dz - 2P)$. With $D = 13$, $P = 18$, $f(z) = 13z^2 - 36z + 18 = r(z)$. □

*Proved here.* That the middle card's lengths stand in this relation is *measured* through its printed agreement with the certificate (Prop. 8.5), and directly in the run of Appendix B, which reads $D = 13$, $P = 18$, $B = 2$ from the fact card's step. $\Phi$ is the operator this card offers toward the map between field spaces that the numerical card says is owed; the relation between the two is taken up in Section 10.

The rest of this section is the dynamics of $G$, which the middle card's readers make depend on how they count. Three facts are needed: when the natural-number pair calculation agrees with the rational one; how $G$ behaves near its fixed points; and what the pairs cost to store.

**Proposition 9.4** (Pairs and ratios)**.** *Let $P \geq D$, and let $p, q$ be positive integers with $z = p/q > P/D$. Then both components of $\Phi(p, q)$ are positive, truncating subtraction and integer subtraction agree in them, and $p'/q' = G(z)$. If instead $z < P/D$, the second component truncates to $q' = 0$ and the pair leaves the rational map.*

*Proof.* $BDpq - BPq^2 = Bq^2(Dz - P) > 0$. Since $P \geq D$, $z > P/D \geq 1$, so $z^2 > z > P/D$ and $Dp^2 - Pq^2 = q^2(Dz^2 - P) > 0$. The ratio is Prop. 9.3. For $z < P/D$, $BDpq < BPq^2$. □

*Proved here.* Both $(13, 18, 2)$ and $(14, 19, 3)$ have $P \geq D$, so the pair calculation follows $G$ along any orbit that stays above the pole $P/D$.

**Proposition 9.5** (Multipliers)**.** *Where $Dz \neq P$,*

$$G'(z) = \frac{D\,(Dz^2 - 2Pz + P)}{B\,(Dz - P)^2}.$$

*(a) If $B = 2$ the numerator is $Df$, so $G'$ vanishes at every fixed point: the fixed points are superattracting, which is the quadratic law of Prop. 8.3.*

*(b) For $(D, P, B) = (14, 19, 3)$ the fixed points solve $28z^2 - 57z + 19 = 0$,*

$$z_\pm = \frac{57 \pm \sqrt{1121}}{56}, \qquad z_+ \approx 1.6157, \quad z_- \approx 0.4200,$$

*with $G'(z_+) \approx -2.0827$, repelling, and $G'(z_-) \approx 0.1494$, attracting. Since $G'(z_+) \neq 0$, $G$ is not Newton's update of any twice-differentiable function with a simple zero at $z_+$.*

*Proof.* The quotient rule gives $G' = \big(2Dz(Dz-P) - D(Dz^2-P)\big)/\big(B(Dz-P)^2\big)$, whose numerator is $D(Dz^2-2Pz+P)$. For (a), $f = Dz^2-2Pz+P$ vanishes at the fixed points when $B=2$. For (b), $(B-1)D = 28$, $BP = 57$, the discriminant is $57^2 - 4\cdot 28\cdot 19 = 1121$, and the multipliers follow by substitution. Newton's update of $f$ has derivative $ff''/f'^2$, which is zero at a simple zero of $f$. □

*Proved here*; the decimals are evaluated from the closed forms.

**Proposition 9.6** (The invariant half-line)**.** *Let $(D,P,B) = (14,19,3)$. On $(19/14,\infty)$ the map $G$ has a single critical point $z_c = (19+\sqrt{95})/14 \approx 2.0533$, a minimum, with*

$$G(z_c) = \frac{19+\sqrt{95}}{21} \approx 1.3689 > \frac{19}{14},$$

*and $G \to +\infty$ at both ends. Hence $G$ maps $(19/14,\infty)$ into $[G(z_c),\infty)$, a subset of itself, and on it:*

*(i) every pair of an orbit begun above the pole has positive components and follows $G$ (Prop. 9.4);*

*(ii) the only fixed point is $z_+$, which repels; no orbit from a rational seed converges to a fixed point;*

*(iii) the halting test can hold only where $|G(z)-z| = \big|28z^2-57z+19\big|\,/\big(3(14z-19)\big) < \tau$, that is, within about $\tau/3.08$ of $z_+$.*

*Proof.* The critical points are the roots of $14z^2-38z+19$, namely $(19\pm\sqrt{95})/14$; the smaller, about 0.661, lies below the pole. On $(19/14, z_c)$ the numerator of $G'$ is negative and beyond $z_c$ positive. At the pole the numerator of $G$ is $95/14 > 0$ and the denominator tends to $0^+$; at infinity $G(z) \sim z/3$. At $z_c$, $Dz_c^2 = 2Pz_c - P$, so $G(z_c) = 2P(z_c-1)/\big(B(Dz_c-P)\big)$ with $Dz_c - P = \sqrt{95}$ and $z_c - 1 = (5+\sqrt{95})/14$, which gives $(19+\sqrt{95})/21$; and $14(19+\sqrt{95}) > 21\cdot 19$ because $\sqrt{95} > 9.5$. For (ii): $z_-$ lies below the pole. An orbit converging to the repelling $z_+$ must equal it eventually; but $1121 = 19\cdot 59$ is not a square, so $z_+$ is irrational, and $G$ maps rationals to rationals. For (iii): $G(z)-z = \big(-(B-1)Dz^2 + BPz - P\big)/\big(B(Dz-P)\big)$ vanishes on the half-line only at $z_+$, where its derivative is $G'(z_+) - 1 \approx -3.08$. □

*Proved here.* For Newton's update $(13,18,2)$ the same half-line above the pole $18/13$ is mapped into $[\alpha,\infty)$ (Prop. 8.3), where $\alpha$ attracts.

Convergence and halting are distinct. By (ii) no orbit of the $(14,19,3)$ step from a rational seed converges; by (iii) such an orbit may still enter the window about $z_+$ at some finite turn, and there the test declares a zero in the sense of Def. 9.2. Whether a given orbit enters the window within a given number of turns is a fact about that orbit, and for the run on the card between it is reported as run evidence (Appendix B).

**Proposition 9.7** (What the pairs cost)**.** *Along an orbit above the pole, $q' = Bq^2(Dz-P)$. For $(14,19,3)$, at every pair after the first, $q' \ge (2\sqrt{95}-19)\,q^2 \approx 0.4936\,q^2$. For $(13,18,2)$, at every pair after the first, $q' \ge 6\sqrt{10}\,q^2$.*

*Proof.* $q' = BDpq - BPq^2 = Bq^2(Dz-P)$. After the first turn, $z \ge G(z_c)$ for $(14,19,3)$ (Prop. 9.6), and $B(D\,G(z_c)-P) = 3\big(14(19+\sqrt{95})/21 - 19\big) = 2\sqrt{95} - 19$. For $(13,18,2)$, $z \ge \alpha$ after the first turn and $2(13\alpha - 18) = 6\sqrt{10}$. □

*Proved here.* The map is homogeneous of degree two, so its pairs are never reduced, and their size roughly squares at every turn whatever the ratio does. This is a cost of the representation, and the two readers pay it alike. From the fact card's start $(530, 324)$, Newton's pairs have 21, 45, 93, 190, 382, 766 and 1535 bits over the seven turns to the certified integer, and there the halting test holds. A pair of the $(14, 19, 3)$ step begun at a three-digit seed such as $(531, 325)$ has about 7.5 million digits at its twenty-first turn; continuing the doubling, about $6 \times 10^{32}$ at its hundred and seventh. *Proved here* for the bound; the bit and digit counts are evaluated exactly for the stated seeds.

**Carriers.** A *carrier* is a symbol, a record called the *slot*, and an interaction (the patter, Definition *Carrier*, in *The table: carriers, marks, tapes, readers*). On a record of more than one symbol the interaction keeps the first symbol and copies the slot beneath it; on a record of one symbol it returns the bare truth $T = T$ in place of the symbol it was handed. It never reads the slot, but no carrier exists without one: the slot is a witness that must exist and is never inspected. By the Quarter (*The quarter is a choice*), a slot is a choice. Read as the axiom of choice, this needs a family, for each carrier the set of its admissible slots, and a principle that selects one slot from every member at once; supplying one explicit finite slot to one carrier is a construction and not an instance of the axiom. In intuitionistic set theory, choice for all such families yields excluded middle [6]; that the middle card's supply of slots is such a selection is *proposed.* The *card between* is a carrier whose cards are themselves carriers of facts; it deals every step of the middle card's ladder identically with the outer cards through the twelfth, and differs from the thirteenth, where the present's quantum is lifted. *Measured.*

**Definition 9.8** (Two readers through a carrier)**.** Fold a tape $t$ onto the slot $q$ of a carrier, giving the record $t\,q$. The *exact reader* reads $|t\,q| - |q|$. The *search reader* reads $|t\,q|$.

**Proposition 9.9** (The exact reading through the card between)**.** *The exact reader reads every tape at its length, whatever the slot. Hence the reading obtained through the card between by the exact reader is the middle card's reading, $x_\alpha$, to every printed place.*

*Proof.* $|t\,q| = |t| + |q|$. The middle card's reading is a function of the lengths of its tapes alone: two readers agree exactly when their lengths do, and re-lettering every cell leaves the reading unchanged (the patter, *Two readers agree exactly when the lengths agree* and *Re-lettering is free*; *measured*). □

*Proved here*, from measured invariance. The machine check of this corollary on the card between is the run of Appendix B, which confirmed it: the exact reader's pair equals the middle card's digit for digit. *Measured.*

**Proposition 9.10** (The search reading carries the quarter)**.** *The search reader reads every tape at its length plus the length of the slot. Its reading differs from the exact reading only through the slot, and agrees with it exactly when the slot is deducted. On a slot of one cell, the inferred step* $(13, 18, 2)$ *becomes* $(14, 19, 3)$.

*Proved here* for the lengths; *measured* for the step, whose three lengths each gain the slot's one cell in the run of Appendix B. The slot's value never enters; only its length does, through the one place the interaction reaches it. The jar, which the middle card builds from its own lengths and not from the reader, is the jar of Prop. 8.2, $[2, 17/8]$ in $z$; by Prop. 9.6 the search reader's orbit ranges over $[G(z_c), \infty)$, and it does not stay in the jar: $G$ maps $[2, 17/8]$ into $[G(z_c),\, G(17/8)]$ with $G(17/8) = 1415/1032 < 2$, below the lower wall. *Proved here.*

*Remark* (The two readers, sealed and observed). Two predictions were sealed before the run on the card between (Appendix B): that the exact reading reproduces $x_\alpha$ to all twenty-four places, and that the search through the outer carrier finds the reading displaced by about one part in $10^7$, about $1.4 \times 10^{-5}$ in reported units, in the same jar.

Outcome (Appendix B). The exact reader reproduced $x_\alpha$, as Prop. 9.9 proves it must. For the search reader three statements are kept apart.

(i) Its step is $(14, 19, 3)$, which by Prop. 9.5 is no longer a Newton update; by Prop. 9.6 its exact orbit converges to no fixed point, and the halting test can hold only in a window of width about $\tau/3.08$ about the repelling $z_+$; by Prop. 9.7 its pairs square in size at every turn. *Proved here.*

(ii) The guarded run returned no reading before its guard stopped it: it stopped at 18 GB after eight minutes against a guard of 16 GB and fifteen minutes, in compiled evaluation on unreduced pairs. *Measured*, as the outcome of the run.

(iii) The stopping condition was not met in the tested sequence: the ratio model of the walk, computed from the measured lengths, ran its 107 turns without entering the window. *Measured* for the lengths; the walk is computed from them.

Statement (i) does not imply (ii) or (iii); they are reported from the run. Carried to its 107 turns, the pair calculation would have required integers of order $10^{32}$ digits (Prop. 9.7).

The sealed displacement assumed a perturbed Newton step, and the search reader through the card between does not take one: the carried slot changes the step itself. That the two readers differ exactly by the slot remains *proved here.* A different search, one that carries the quarter without breaking the derivative, would be a new construction and would need a new sealed prediction.

For scale, as the prediction stood when sealed: a displacement of $10^{-7}$ is about eighteen hundred times smaller than the gap between $x_\alpha$ and the recommended value (Section 11). It was also set beside the shift of $2.1 \times 10^{-7}$ that the reading induces in the predicted anomalous moment (numerical card, *The comparison with the measured constant*); the displacement is relative, in the reading, and the shift absolute, in the moment, so the two compare only through a calculation of how the moment responds to the reading.

**The search for the quarter.** To find the slot is to range over carriers: every record is a valid slot, and each makes the carrier exist. Three things are kept apart. *The guard*: the middle card's heartbeat meter halts a dealing loop when the work of a program exceeds four times a reference plus a measured noise floor, and its calibration treats a stop at the checker's runtime limit as an unsuccessful attempt; the source names the four as the fourth busy beaver. This is a stopping policy the card defines, not a computability result. *Measured. The observed stop*: the search reader's run of Appendix B was stopped by its memory guard, not by the meter. *Measured. The correspondence*: the fifth busy beaver [16] concerns the halting of five-state, two-symbol Turing machines, a different machine model; that what lies past the meter's bound is its territory is *proposed*, and the name is kept as the bound's name. Herbrand's theorem [12] likewise concerns a first-order language: a sentence, after Skolemization, is unsatisfiable exactly when some finite set of its ground instances is propositionally unsatisfiable, so a refutation search ranges over ground instances. That the search for a slot ranges over such a space needs a first-order language for slots and a procedure over its ground instances; it is *proposed.* Within the search, the forced universe levels fix the value and the

free ones carry the cost: moving every free universe seat leaves the reading unchanged (Appendix B). *Measured.*

**Definition 9.11** (Covariant and contravariant comparison)**.** Two marked wholes $(f_1, n_1)$ and $(f_2, n_2)$ compare by

$$(f_1 = f_2 \ \wedge \ n_1 \leq n_2) \ \vee \ (f_1 \neq f_2 \ \wedge \ n_2 \leq n_1).$$

The first branch is *covariant*: the marks agree and the order runs forward. The second is *contravariant*: the marks differ and the order reverses. Covariance is $X = X$; contravariance is $X = Y$.

*Measured*: this is the middle card's first order, on its naturals and again on its rationals, where the contravariant branch is called the missing branch. Its companion order on carried values names the covariant true values as those that close by reflexivity and the contravariant ones as those that do not.

*Remark* (Two readings of one chain)*.* Take the chain of the card between: the fact that $T = F$ is $F = T$, the truth $T$, and the fact that $T = T$ is $F = F$.

*Covariantly*, read by how each fact clicks, the three click alike, the chain closes in one class, and the closure is a check the apparatus accepts, billed as a value identity. On this reading the chain is a proof.

*Contravariantly*, read by what each fact says, the three are not one by reflexivity; the comparison takes the second branch of Def. 9.11 and the order reverses. On this reading the same chain is the line an indirect proof writes under its hypothesis: a counterfactual that must be writable so that a proof of its non-existence can be completed, and that never clicks.

The two tables, by meaning and by click, agree cell for cell and differ only in which diagonal clicks (the patter, *Same difference*, in *The throw: what is preserved, and the rule*). So the two readings carry the same content in opposite orientations, and the card between is where they meet.

*Measured*: the closure of the chain by click and its value-identity bill; the absence of reflexive identity among the three truths on the outer cards; the cell-for-cell agreement of the two tables. *Proposed*: that the covariant reading is a proof of the identification of Section 10, and the contravariant reading the counterfactual its refutation requires. The proposal cannot become a step until that identification is a statement the apparatus can hold; a search that found it by refuting its negation would be an indirect proof, and would bill choice, which the reading of this card never does.

## 10 Electrodynamics as the derivative

The numerical card (*The derivative of the modified field equations*) proposes an identity between linearized operators. Let $X_{\mathrm{QED}}$ and $X_{\mathrm{GR}}$ be the field spaces of its bare electrodynamics and its first-order gravity, with their declared norms, so that $R_{\mathrm{QED}} : X_{\mathrm{QED}} \to X^*_{\mathrm{QED}}$ and $DR_{\mathrm{GR}}(g) : X_{\mathrm{GR}} \to X^*_{\mathrm{GR}}$. The proposal is a background $\psi_0 \in X_{\mathrm{QED}}$ and a differentiable map $\Psi : X_{\mathrm{QED}} \supset U \to X_{\mathrm{GR}}$, with residual values compared by the pullback $D\Psi(\psi_0)^* : X^*_{\mathrm{GR}} \to X^*_{\mathrm{QED}}$, for which

$$DR_{\mathrm{QED}}(\psi_0) = D\Psi(\psi_0)^* \circ DR_{\mathrm{GR}}\big(\Psi(\psi_0)\big) \circ D\Psi(\psi_0) \qquad \text{on } X_{\mathrm{QED}}. \tag{10.1}$$

*Proposed*, as on the numerical card, with electrodynamics the theory of Dirac's electron [7]. The numerical card writes $\Phi$ for the map; here $\Phi$ is the inferred operator of Def. 9.1, and the field map

is written $\Psi$. Each side is linear in the perturbation at its fixed background; dependence on the background is a separate property and is not asserted.

The middle factor of (10.1) is the linearized operator of the first-order variables $(h, \Gamma)$. This card's operator is the metric one, reached by the passage (i)–(iii) of Section 6: change of variables, elimination of $\Gamma$ by its stationary equation, restriction to the unimodular sector. At a background $g$ with $E(g) = 0$ and $R(g) = 4\Lambda$, Prop. 6.3(b) gives that operator on the constrained sector,

$$k \;\longmapsto\; DE_g[k] + \Lambda k, \qquad DR_g[k] = 0,$$

with the affine term contributing the multiplier and nothing else. Read in metric form, therefore, (10.1) takes the middle factor to be $DE_g + \Lambda$, and it requires the range of the metric part of $D\Psi(\psi_0)$ to lie in the constrained sector; on that range the multiplier contributes $\Lambda\, D\Psi(\psi_0)^* D\Psi(\psi_0)$, with residual values paired with perturbations through $g$. For variations outside the sector the term $\frac{1}{4} DR_g[k]\, g$ of case (a) remains, and the identity would have to account for it. The passage (i)–(iii) relates the equations; that it relates their linearizations, so that the first-order middle factor may be replaced by the metric one, is a further statement, which this card does not carry out. *Proved here* for the form of the middle factor on the constrained sector; the identity itself stays *proposed*.

$\Phi$ of Def. 9.1 is a map on pairs of integers; $\Psi$ is a map between field spaces. No relation between them (an embedding, a discretization or a restriction) is stated on this card; that $\Phi$ is the scalar shadow of $\Psi$ is *proposed*.

**What this card supplies.**

(a) The shape, in the scalar case, in full: a cubic action, its residual, the residual's derivative with remainder exactly $13h^2$, and at the reading the derivative squaring to the discriminant (Prop. 7.1). *Proved here.*

(b) The operator: $\Phi$ of Def. 9.1, the step the middle card infers, which for $(13, 18, 2)$ is Newton's update and carries the first variation to its zero, the fixed point of the update (Props. 9.3, 8.3, 8.5). *Proved here* and *measured*.

(c) The agreement: the search by the first variation and the search by the derivative share a jar and a point (Section 8). *Proved here* and *measured*.

(d) The carrier: read through the card between, the exact reading is unchanged (Prop. 9.9), and the search reader's change of the calculation is exact (Props. 9.5 and 9.6). *Proved here*, and *measured* in the run of Appendix B.

(e) A finite piece of the gauge-and-spinor side, written from the device's own tables (the argument card, Section 9): on a torus of any size, links that are quaternions with whole entries, an action on each square that is exact at the second order and invariant when every site is turned by a link of length one, flat configurations stationary on every link, real Dirac matrices, and on a box of eight a phase every gamma commutes with, a train line whose current makes the derivative, and the whole action of light and matter gauge invariant for every weight $\beta$ between them, in two directions and in four. *Measured*. It is a family of tori, each on its own, not a field space, and it does not supply $\Psi$.

**What remains a sacred text.** That a map $\Psi$ from the gauge-and-spinor field space to the gravitational one exists, related to $\Phi$ by a stated embedding, discretization or restriction; that the identity (10.1) holds at a stated background with the perturbation and residual spaces matched; and, for the quantum theory, the regulator the numerical card requires (*Bare QED on a fixed background*). These are the three items the numerical card lists as owed, and this card does not discharge them.

**Pointwise exact, in the only sense the norm allows.** The identification wants position and velocity exact in the Sobolev norm of first order. That norm controls both in mean square. It gives no point values of the velocity, and in two or more dimensions none of the position either, so a first-order Sobolev statement does not supply the derivative values a Hermite interpolant needs; those must come from a smoother solution. Where they are supplied, exactness is taken where it can be had: at knots. A Hermite cubic matches position and velocity at every knot, and on $N$ equal cells of $[a, b]$ it differs from any $\phi$ with a bounded fourth derivative by at most $\frac{(b-a)^4}{384N^4} \left\|\phi^{(4)}\right\|_\infty$, so each halving of the cells divides the bound by sixteen (cheat sheet, *Step one: the difference between the model and the device*). *Standard*, proved there. Every continuous function on a closed interval is a uniform limit of polynomials [18, 4], and smooth functions are dense in the Sobolev space [13]. *Standard.* Uniform convergence is convergence of values, not of derivatives, and so not of the stationary equations, which involve derivatives; approximation in the first-order Sobolev norm is what the density supplies. So "pointwise exact" means exact on a ladder of knots, with the solution trapped between splines and the ladder converging in norm; that the device's run, never finished, contains the limit is *proposed*. What the ladder does not supply is the remaining item: that a single cubic action has one stationary point that is the principle of stationary action for both the gravitational and the quantum model. The numerical card proves the two share one derivative hierarchy (Thm. 2.4); sharing a hierarchy is not having one action, and the step between them is the map $\Psi$ above. It is owed. The literature's standard statement, that gravitational amplitudes are squares of gauge amplitudes [1, 2, 3], has the shape of the proposal: the derivative of a square is linear in its root. It is Pythagorean here, used for that shape and nothing else. The identification is *proposed*.

## 11 The record

The publication carries a register, a counter, a page trace and a verdict.

**Register.** The constant $\Lambda$ of Prop. 6.1: fixed by data, carried unchanged by every solution of fixed integration constant (Prop. 6.3(c)). The reading $x_\alpha$ is entered beside it as the register's representative, which by the Quarter is a choice.

**Counter.** The halvings of Prop. 8.4, strictly increasing from 0 to 108 on this card's budget.

**Trace.** At each count $n$ the bracket $[L_n, U_n]$, and beside it the derivative's iterate when one exists. The trace's head is the jar.

**The press check.** Two triples are read from the one register.

**Proposition 11.1** (The two verdicts)**.** First variation against derivative. *Slipped: false. Value: the certified integer* 13701129054897945708773730 0 *at scale* $10^{24}$*. Velocity:* 108 *halvings against* 5 *turns.*

Exact reader against search reader, through the card between. *Slipped: true exactly when the slot is not deducted. Value: the exact reader's $x_\alpha$; for the search reader, no reading. Velocity: the length of the slot, one cell on each of $D$, $P$ and $B$.*

*Proved here* and *measured* for the first triple. For the second, *proved here* that it slips exactly through the slot and that the slip changes the step from $(13, 18, 2)$ to $(14, 19, 3)$, whose exact orbit converges to no fixed point (Prop. 9.6). The search reader's entry in the value is *measured*, from the run of Appendix B: the run returned no reading before its guard stopped it, and the stopping condition was not met in the tested sequence (Section 9). It is a report of that run, not a consequence of the proposition. The sealed *prediction* that the reading slips by about one part in $10^7$ is not borne out. One register yields one triple that does not slip and one that slips by the quarter, which is the shape of the numerical card's own press check.

**Against the world.** The inverse fine-structure constant has the recommended value [14]

$$137.035999177(21),$$

and the reading lies 1.80 parts in ten thousand below it. Twenty-four places of agreement between the two searches show that the gap is not numerical noise: the searches hit their own number exactly, and the residue belongs to the description, not to the computation. The general claim that the reading is the fine-structure constant is excluded by the measured anomalous moment [9] (numerical card, *The comparison with the measured constant*); only the narrow form, a prediction for one stated arrangement (cheat sheet, *Two revolutions*), is offered. *Triangle*, in the sense of Def. 1.1: this is the comparison that can slip, and it does.

**And what is this, and what is that.** The reading is a ratio of counts and carries no unit. Calibrate the clock: one symbol is one tick, and one tick is one period of the caesium hyperfine line, 1/9 192 631 770 s, about 108.78 ps. This is a stipulated calibration: the record counts symbols, and the second is assigned to them. The only length the record then has is $c\,\Delta t$ for one tick, about 3.2612 cm, a length assigned through that calibration (it is the line's own wavelength), so the speed of light is one symbol per tick and 299 792 458 m/s returns as the ratio of the two SI definitions, the metre to the second. *Standard*, and a convention: the speed of light has been exact since 1983, and the Planck constant and the elementary charge were fixed exactly in the revised SI of 2019 [5]. Calibrate next to the reduced Planck constant and keep the SI convention that the elementary charge is exact [14]. With $c$, $\hbar$ and $e$ all declared, the one electromagnetic constant left to measure is fixed by the reading alone, through $\mu_0 = 2\alpha h/(ce^2)$ and $\varepsilon_0 = 1/(\mu_0 c^2)$:

| | $\mu_0$ (H/m) | $\varepsilon_0$ (F/m) | $Z_0 = \sqrt{\mu_0/\varepsilon_0}$ ($\Omega$) |
|---|---|---|---|
| recommended $\alpha$ | $1.256\,637\,061 \times 10^{-6}$ | $8.854\,187\,82 \times 10^{-12}$ | 376.730 313 |
| the reading | $1.256\,863\,683 \times 10^{-6}$ | $8.852\,591\,34 \times 10^{-12}$ | 376.798 253 |

The columns are not independent tests. With $c$, $h$ and $e$ fixed, each of $\mu_0$, $\varepsilon_0$ and $Z_0 = \mu_0 c$ is a fixed function of $\alpha$, so their recommended values share one correlated uncertainty, that of $\alpha$, and the table is one comparison written in several expressions. *And what is this*: the product. $1/\sqrt{\mu_0\varepsilon_0}$ is

299 792 458 m/s from either row; the reading leaves the speed of light exactly where the convention put it, because it moves the permeability up and the permittivity down by one factor. *And what is that*: the ratio. The impedance of free space, $Z_0 = 2\alpha h/e^2$, is where the whole of the reading's difference lands: the vacuum resists a wave 1.80 parts in ten thousand more than recommended. The speed is the declared constraint and the impedance its measured price, the multiplier and the penalty of Section 6. *Proved here* as arithmetic from the reading and the declared constants; that the speed is the multiplier and the impedance its price is *proposed*. Under the other common convention, the permittivity declared one, the same difference appears instead as an elementary charge 0.9 parts in ten thousand small. The convention chooses where the residue is written, never whether it is there. The impedance, known to about one part in $10^{10}$ because $\alpha$ is, excludes the reading; this is the comparison above written once more, not a second exclusion beside the anomalous moment's.

## 12 What the knower carries

Collected, the proposals of this card are five sentences. The search by the first variation is the principle of stationary (least) action. The search by the derivative is electrodynamics. The inferred operator is the scalar shadow of the map between the field spaces. What lies past the meter's bound is the fifth busy beaver. The search carries the quarter. None is a step; each is about the continuation of a record of which this card has only a prefix. The fifth was sealed with a weight, one part in ten million, and the run did not bear it out (Prop. 11.1).

The prefix is the propositions of Part II under their calibrations, and its received texts enter by the rule of admission (Prop. 1.2) at the places the canon names. A reader who accepts every one of them is still handed the five sentences calibrated *proposed*, so that they are carried as texts and not mistaken for receipts.

The argument closes on the bent card (*The Infinite Loop Considered Harmless*, *Three bent cards*). The record needs three receipts it cannot write for itself: the base case of its search, which Prop. 9.9 supplies; the zero its halting test declares; and the constants that give it units, under which the reading becomes the impedance of free space. Each is a fixed point handed in from outside, and none absorbs the residue.

**The last sacred text.** The canon of Section 2 lists the texts the derivation stands on, and Section 4 follows them down to the electron. One more is not on either list, because the device cannot cite it: it is the text the reader brings.

The local result is the second verdict of Prop. 11.1. Beside it, every reader that measures length, every slot, every free seat, both cuts, and two machines leave the reading where it is, and no declaration carries $x_\alpha$ onto the recommended 137.035999177(21); the residue of $1.80 \times 10^{-4}$ stays. *Measured*.

Three statements are kept apart.

(i) A *bounded unsuccessful search*: these constructions, run under these guards, do not carry $x_\alpha$ within the measured uncertainty of the recommended value. That is what the record shows.

(ii) A *universal claim*: no construction in a stated class of carriers, readers and declarations does so, $\forall x \in C\ \neg P(x)$, equivalently $\neg\exists x \in C\ P(x)$. It quantifies over the class and needs a proof

over it. Passing to it from (i) is a generalization beyond the cases searched; no logical principle licenses that passage, and no finite search supplies the proof.

(iii) *Double-negation elimination*, the logical principle $\neg\neg A \to A$. Applied with $A = \exists x \in C\ P(x)$, it passes from a refutation of (ii) to the existence of a construction without exhibiting one. It is valid classically and not intuitionistically; it is not the passage from (i) to (ii).

The passage from (i) to (ii) has the shape of the halting test's declared zero (Def. 9.2): a finite stop read as the limit it stands for. It is taken by belief.

So the last sacred text is a sentence and the belief that travels with it: *the device's failure to construct a reading within the measured uncertainty of* 137.035999177(21) *demonstrates that there is none.* It is Pythagorean in the sense of Def. 1.1, admitted at one place only, the residue, and never shown. The reader's question is the Quarter's: can you find the polynomial in the physical model of this device? The knower's choice is whether this text is true. The rest of this card is of other kinds, each with its calibration: derivations inspected on the card, results cited, runs reported, definitions stipulated, and correspondences proposed.

■

# A Sources

This appendix is the only part of the card that names the middle card's files and declarations. Paths are relative to `CARD_2/Measurement/`; declarations are cited by name. The working tree was inspected on 31 October 2026, and again for the inferred step on 3 November 2026; this appendix reports no build.

| This card | Middle card (declaration) |
|---|---|
| Def. 1.1 | `Episode04.lean`: `Cult` (`inside_joke`, `pythagoras`, `triangles`), `InitiationProcess.sacred_texts`; `Episode09.lean`: `STEP25_BUILD_LOCAL_GEOMETRIES_THAT_SATISFY_THE_TRIANGLE_INEQUALITY`; `Episode10.lean`: `REVIEWER_2_ASKS_IF_DEVOTION_TO_METHOD_IS_A_METHOD` |
| Epigraph | `Episode04.lean`: margin of `the_experiment` |
| §3 | `Episode10.lean`: `REVIEWED`; `Episode11.lean`: `TYPESET`, `pressCheck` |
| §4, work count | `Calibration/LeanCalibration.lean`: `register_noise_floor`, `measuredFloor`; toolchain `leanprover/lean4:v4.27.0-rc1` |
| Prop. 8.2 | `Episode12.lean`: `jarLo`, `jarHi`, `theLowWall`, `theHighWall` |
| Prop. 8.5 | `Episode12.lean`: `halve`, `frechet_walk`, `theBisectionLedger`, `theNewtonLedger`, `the_bisection_ledger_turns`, `the_newton_ledger_turns`, `ledger_is_the_fold`, `the_crank_is_its_ledger`, `the_bracket_is_its_ledger`, `the_low_wall_is_above_the_vertex`, `the_printed_reading_is_the_natural`, `the_division_was_quot_sound`, `what_would_you_call_this_number?` |
| Def. 9.1; Props. 9.4–9.7 | `Episode08.lean`: `bent_card.the_device_computing_the_next_configuration` (the components `x'`, `y'` and the halting test on `Nat` with truncating subtraction), `bent_card.fencepost`, `bent_card.the_exact_Taylor_series_degree`, `bent_card.the_length_of_the_path_from_here_to_there`, `bent_card.the_measurement_degrees_of_freedom`, `infer`, `the_truth_inferred_by_the_device` |

| This card | Middle card (declaration) |
|---|---|
| Carriers | `Episode01.lean`: `CarrierProcess` (`symbol`, `quarter`, `interact_with_the_universe`), `FACT.Truth`; `Episode08.lean`: `Number.peano`; `Episode12.lean`: `the_card_between`, `the_mark_between`, `the_tape_between_deals_rung01`, `deal_rung`, `upness_seats` |
| §10, pointwise exact | `Episode03.lean`: `Polynomial.le` and `Spline` with their Weierstrass margins; `Episode04.lean`: `Variation.le`, the `the_frechet` arm and its margin (same position, same velocity, same acceleration) |
| Def. 9.11; two readings | `Episode01.lean`: `Natural.le` and the rational order with its covariant/contravariant margin; `Episode02.lean`: `PigeonHole` order (superpigeons, pigeons, boxes); `Episode12.lean`: `same_click`, `the_chain_closes`, `THE_ACCOUNT_CLICKS`, `the_closure_between`, `SAME_DIFFERENCE`, `the_click_diagonal` |
| Prop. 9.9 | `Episode12.lean`: `the_other_machine_agrees`, `relettering_is_free`, `the_clicks_read_the_number` |
| The search for the quarter | `Calibration/LeanCalibration.lean`: `EKG`, `EKG.outgrown?`, `busyBeaverBudget`, `register_noise_floor`; `Calibration/EKGBounded.lean`: `tryBounded?` (a runtime-limit stop returns `none`); `Episode05.lean`: `teraFLOPs`, `trilinos`, `HALTED` |
| §11 | `Episode11.lean`: `TYPESET`, `PressVerdict`, `pressCheck` |

# B The sealed predictions

Written on 31 October 2026, before the run that decides them: the middle card built from its working tree on a second machine, then a three-reader comparison through the card between (the carrier's own interaction, the search reader, and the exact reader of Def. 9.8).

- The exact reader reproduces 137.011290548979457087737300 to every place, bills symbol identity only, and agrees by definitional unfolding with the middle card's pair.
- The search reader finds the reading displaced by about one part in $10^7$.
- Both lie in the jar $[129.6, 137.7]$.
- The carrier's own interaction, applied to a tape folded onto a zero, keeps its first symbol and the slot, so every nonempty tape reads two cells and the reading it feeds the search is degenerate.

The predictions above are left as sealed. The run's outcome is entered below, and the body cites it where it uses a prediction (Section 9, Prop. 11.1, Section 12).

**Outcome, 31 October 2026.** The middle card built cleanly on the second machine (twenty-two minutes; peak resident memory 1.56 GB). The three-reader comparison was then started on one core and stopped by hand after seventy minutes, with its memory footprint at 92 GB on a machine of 64 GB and the excess in compressed memory and swap. It printed no reading. A first attempt on a machine of 8 GB had been stopped at 7 GB for the same reason. *All four predictions above remained undecided by this first run.* The first was not thereby weakened: it follows from Prop. 9.9 on paper, and the run was its machine check, not its proof. Separately, on the same day and machine, every free universe seat of the ladder's rungs fourteen to thirty-eight was moved off its computed pin;

all twenty-five rungs dealt, clicked exactly as the pinned rungs do, and billed the same receipts. *Measured.*

**Second run, the same day.** The comparison was repeated in four stages, each guarded at 16 GB and fifteen minutes, with the walk read from where the fact card stores it instead of being recomputed.

- *Exact reader: confirmed.* Its pair equals the middle card's pair digit for digit, and its reading is 13701129054897945708773730 0 at scale $10^{24}$, the device's own, in under a second. The receipt is proved: folding a tape onto a slot appends the slot (no axioms); the exact reader is the length (symbol identity); the reading through the card between is the middle card's (symbol and value identity, the latter from function extensionality, so the receipt costs one value identity more than the first line above predicted; the reading itself costs none).
- *Carrier's own interaction: confirmed.* Its pair is $(0, 0)$: every tape reads two cells, the start lands on a double root, and the first step sends it to zero.
- *Search reader: no reading.* Stopped by the guard at 18 GB after eight minutes, in compiled evaluation. The measured lengths explain it. The fact card's step has $D = 13$, $P = 18$, $B = 2$ and starts at $(530, 324)$; this is $N$ exactly (Prop. 9.3), and on ratios it halts at its seventh turn on the certified integer, which a ratio model of the walk reproduces to every place. The search reader adds the slot's one cell to every length: $D = 14$, $P = 19$, $B = 3$, and the step no longer divides by the derivative. Its fixed points are $(57 \pm \sqrt{1121})/56$; the upper one, the counterpart of the reading, repels, with multiplier about $-2.08$ (Prop. 9.5). The unnormalized pair squares its size every turn (Prop. 9.7), and the run exceeded its 16 GB guard after eight minutes, before any reading; the hundred and seven turns would have required integers of order $10^{32}$ digits. In a ratio model of the walk the halting test never holds in its 107 turns, and the walk wanders and ends at a reported value near 220.10, outside the jar.

*Measured* for the lengths, the pairs and the stopped run; *computed*, from the measured lengths, for the ratio model. So the first prediction holds, the fourth holds, the third holds only in the sense that the jar is built from the middle card's own lengths (the search reader's walk leaves it, as Prop. 9.6 requires), and the second, a displacement of one part in $10^7$, is not what the search reader through the card between does: the run returned no reading before its guard, the ratio model met no stopping condition in its 107 turns, and the carried slot turns the derivative's step into one whose fixed point repels. Whether some other search, one that carries the quarter without breaking the derivative, displaces the reading by that amount is open, and would need a new sealed prediction.

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